

PHY202: 2nd Mid-Semester Exam

Duration: 60 minutes; Maximum marks: 20

1. (a) Find the Fourier series for the periodic function $f(\theta)$ defined as,

$$f(\theta) = |\sin \theta|, \quad -\pi \leq \theta < \pi, \quad \text{and} \quad f(\theta + 2\pi) = f(\theta).$$

- (b) Use the result of part (a) to find the sums,

$$(i) \sum_{n=1}^{\infty} \frac{1}{(4n^2 - 1)}, \quad (ii) \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2(2n-1)^2}$$

[4+2+2]

2. Fourier series for a periodic triangular wave defined as, $f(x) = \begin{cases} -x, & -\pi < x < 0, \\ x, & 0 < x < \pi, \end{cases}$ is given by,

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots} \frac{\cos(n\pi x)}{n^2}.$$

- (a) Draw the periodic triangular wave indicating the axes scales.

- (b) Find the Fourier series for the periodic square wave: $g(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$

[1+3]

3. Fourier transform of a function $f(x)$ and the corresponding inverse transform are defined as,

$$g(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx,$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(k) e^{ikx} dk.$$

- (a) Find the Fourier transform of the function, $f(x) = e^{-a|x|}$.

- (b) Find the inverse transform of the Fourier transform obtained in part (a).

[4+4]